

1 Experimental Results and Discussion

This section consolidates the empirical and simulated findings of the thesis, structured to provide a comparative analysis of the Over-the-Air (OTA) computation performance across different modeling layers. The discussion first examines the degradation mechanics revealed by MATLAB and Simulink simulations, evaluating the efficacy of applied equalization techniques. It then outlines the evaluation framework for the physical Software-Defined Radio (SDR) deployments.

1.a Comparative Analysis of Channel Impairments (MATLAB vs. Simulink)

The rigorous simulation of Over-the-Air (OTA) architectures across discrete-time algorithmic environments (MATLAB) and continuous-time dynamical environments (Simulink) provided deep insights into the physical limitations of signal superposition.

Under ideal Additive White Gaussian Noise (AWGN) conditions, both simulation environments confirmed the theoretical Mean Squared Error (MSE) scaling laws, demonstrating that computation variance is strictly bounded by the channel noise [1]. However, the introduction of uncoordinated physical impairments revealed significant divergence from the idealized theory:

- **Multipath and Fractional Delay:** The high-fidelity simulations demonstrated that even sub-symbol fractional delays (modeled via Farrow filters) induce severe Intersymbol Interference (ISI). The 2D and 3D MSE surface scans revealed an irreducible error floor that scales with the multipath gain, fundamentally bottlenecking the computation accuracy regardless of the transmission power.
- **Asynchronous Carrier Frequency Offset (CFO):** Independent local oscillator offsets among distributed transmitters were shown to convert stationary baseband values into beat-frequency harmonics. Simulink evaluations demonstrated that traditional coherent detection fails catastrophically under these conditions. The application of non-coherent Euclidean magnitude detection mitigated the zero-crossing signal collapse but introduced a multi-user vector fading penalty.

1.b Evaluation of Equalization and Recovery Strategies

The simulations also served to benchmark classical equalization and synchronization recovery schemes. The divergence between algorithmic MATLAB models (which assume perfect block-level knowledge) and Simulink models (which emulate real-time causal processing) highlighted the difficulty of correcting asynchronous CFO post-superposition. While techniques such as squaring loops and Costas loops are standard for point-to-point links, their application to aggregated OTA signals proved inadequate when phase offsets are uncorrelated across multiple transmitters. This finding justifies the ongoing investigation into more robust, adaptive filtering mechanisms.

1.c Framework for SDR Experimental Validation

Building upon the simulation baseline, the physical deployment utilizing GNU Radio and Software Defined Radio (SDR) hardware aims to validate the theoretical error floors in a real-world electromagnetic environment. The planned experimental results will be benchmarked against the Simulink diagnostics to quantify hardware-specific non-linearities, such as amplifier saturation and discrete sampling jitter. Ultimately, the experimental framework will demonstrate the practical computation rate achieved by the OTA approach relative to traditional orthogonal access schemes (e.g., Time-Division Multiple Access (TDMA)), illustrating the critical balance between resource savings and analog computation error.

References

- [1] A. Şahin and R. Yang, "A survey on over-the-air computation," *IEEE Communications Surveys & Tutorials*, vol. 25, no. 3, pp. 1877–1908, 2023, ISSN: 1553-877X, 2373-745X. DOI: 10.1109/COMST.2023.3264649 Accessed: Dec. 4, 2025. [Online]. Available: <https://ieeexplore.ieee.org/document/10092857/>